Supply Chain Management

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Fundamentals of Data Science

Midterm Exam Supply Chain Management

Introduction

This white paper seeks to use the Cross Industry Standard Process for Data Mining (CRISP-DM). This paper will look at the problems that are experienced by supply chain management as it pertains to how data science, data mining, and data modeling can be used for solving various problems that are experienced by the industry.

About the CRISP-DM Model

A brief overview of the CRISP-DM is in order to help give more understanding to what this paper will try to accomplish. The CRISP-DM is a process model that is a framework for a data science activity (Hotz, 2023).

This model is broken up into six sequential phases (Hotz, 2023):

1. Business understanding – the business need.
2. Data understanding - What data does the organization have, what does it need, and is the data clean?
3. Data preparation - Organizing the data for processing.
4. Modeling – What is the best model to answer the question that is trying to be answered?
5. Evaluation – What is the best model to meet the objectives of the business?
6. Deployment – How do we disseminate the results of the model to the interested parties.

This paper will focus on a few areas that were stated above. It will look at business understanding, data discovery and understanding, along with a posture statement. This last section will state what the models will hope to accomplish in regards to the business problem that will be discussed in the following section.

Business Understanding

In order to use the framework proposed by the CRISP-DM model, we must understand the industry that will be applied to. The industry that this paper will look at is supply chain management. In order to figure out the best approach to data mining, as it regards this industry, we need to understand the industry. We first need to understand what it means by the term supply chain management. Supply chain management in its most basic definition is, "handling of the entire production flow of a good or service" (*What Is Supply Chain Management? | IBM*, n.d.).

This industry has many problems associated with it and its use of data, specifically big data. These issues arise in company operations, cost reduction, forecasting demand for products, predictive maintenance, product pricing, and others (*Top 8 Data Science Case Studies for Data Science Enthusiasts*, n.d.).

Data science is key to understanding items like forecasting demands and supply of inventory. This is a major problem that can be overcome using data analytics and its ability to predict outcomes based on historical factors. Data science can look at the types of carriers and predict the likelihood of them being used or they can look at the delays in shipments of products or services (Data Scientists in Supply Chain Management, 2022). The latter example is the key focus of this paper, but the model that will be discussed in the subsequent section can be applied to others if the right inputs are used and the data is of a good quality.

Data science and its role in supply chain management is one that can be a tool that offers insight into what are the causes of the problems as well as give a glimpse into the future. Data science can be used in multiple ways to aid in the management of the supply chain. This paper seeks to address the problem of late delivery of items to the consumer and its possible causes as they relate to the scheduled delivery of the product.

How can data science benefit supply chain management? Data science can improve the efficiency of the processes that are being used by the organization. This efficiency can arise from looking into making the shipping more efficient by looking at the root-cause of any pending problems. These are not the only areas that data science can aid in. The application of data science can aid in manufacturing by ensuring that the right products are made and delivered at the right time (Dunham, 2022).

Using concepts like predictive analytics, diagnostic analytics, or predictive analytics to give a view into the what and the why of an event that has happened to the organization (Burns, 2019). For example, what caused a delay in shipment of an item to the consumer. What were the root causes of this delay? Is it something that was internal to the organization or was it an external to the organization, e.g, logistics problem with the carrier.

The dataset that will be used will look at the delivery methods versus delay in receiving the package as viewed by the customer. The dataset will be discussed in detail as well as the model that will help predict the likelihood of a product being delivered late to the consumer.

Data Discovery and Understanding

The dataset used can be found on the website Kaggle. The full link to the dataset and the card with descriptive information can be found in the footnote below[1]. The dataset contains information about shipping, the customer, the method of shipping, and other data (DataCo SMART SUPPLY CHAIN FOR BIG DATA ANALYSIS, 2019).

In regards to CRISP-DM, the preceding would correspond to collecting the data. The data was reviewed to see what it contains. It was verified for quality and there found to be view rows that did not contain any values and they were dropped from the dataset/dataframe. This did not pose a dataset that was quite large and the few rows that were dropped should not have much of an affect on the quality of the model.

To develop the model we need to understand the hypothetical problem. The problem in this case was the delay of packages being delivered to the consumer. The main features that were examined are Days for shipping (real), Days for shipping (real), Days for shipment (scheduled), Customer State, Delivery Status, and Shipping Mode.

These features were used to devise a K-nearest neighbor model that does an adequate job of predicting the likelihood of a package being delivered late. Other features of the dataset were dropped and were not used to create this model. A link to the full Jupyter Notebook can be found in the footnote below[2]. The reason that this model was chosen is that it is a very simple algorithm that can be easy to implement and easy to change by adding or subtracting features.

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[1] Link to the Kaggle dataset: https://www.kaggle.com/datasets/shashwatwork/dataco-smart-supply-chain-for-big-data-analysis. The title of the set is *DataCo SMART SUPPLY CHAIN FOR BIG DATA ANALYSIS*

[2] Link to the Jupyter Notebook: https://github.com/meheino77/Fundamentals_of_Data_Science-/tree/main/Exams/Midterm

This allows the user of the algorithm to make a wide assortment of predictions with very little knowledge of the mechanics of the algorithm. An artificial neural network was also tried but for simplicity this model was chosen and more than adequate for the purpose of making these predictions and for other data mining processes.

While this is a good algorithm for this particular situation, there is one caveat, it does not scale well with large data sets (Burns, 2019). This can be overcome by taking a suitable sample that has been randomized and used in the model.

The results of the algorithm was that its accuracy was approximately 95%. This was with data that was not the best as it did not have all the information. This could lead to an organization seeking to collect further information to help narrow down the causes of the delays. The organization could look to collect information about possible incidents that may have affected the delivery of the product to the consumer. For example, was there an unexpected weather delay or was there a delay in procuring the necessary materials to deliver the product within the timeframe specified. This type of information is outside the scope of the dataset. But as it pertains to the second phase of the CRISP-DM model it does force the organization to see if they have other data that can be used to help illuminate the results of the model. This allows the organization to gain a better perspective on the results of the model..

Brief Summary of Model Creation: The model was created in the usual manner. The dataset was read from the CSV named, “DataCoSupplyChainDataset.csv”.

	Type	Days for shipping (real)	Days for shipment (scheduled)	Benefit per order	Sales per customer	Delivery Status	Late_delivery_risk	Category Id	Category Name	Customer City	Customer Country	
0	DEBIT	3	4	91.250000	314.640015	Advance shipping	0	73	Sporting Goods	Caguas	Puerto Rico	X
1	TRANSFER	5	4	-249.089996	311.359985	Late delivery	1	73	Sporting Goods	Caguas	Puerto Rico	X
2	CASH	4	4	-247.779999	309.720001	Shipping on time	0	73	Sporting Goods	San Jose	EE. UU.	X
3	DEBIT	3	4	22.860001	304.809998	Advance shipping	0	73	Sporting Goods	Los Angeles	EE. UU.	X
4	PAYMENT	2	4	134.210007	298.250000	Advance shipping	0	73	Sporting Goods	Caguas	Puerto Rico	X

Figure 1. Dataframe of the DataCo CSV

Gathered information about the columns and the data type of each columns

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 180519 entries, 0 to 180518
Data columns (total 13 columns):
 #   Column                         Non-Null Count   Dtype
---  ------                         --------------   -----
 0   Days for shipping (real)       180519 non-null  int64
 1   Days for shipment (scheduled)  180519 non-null  int64
 2   Delivery Status                180519 non-null  object
 3   Late_delivery_risk             180519 non-null  int64
 4   Customer State                 180519 non-null  object
 5   Customer Zipcode               180516 non-null  float64
 6   Order Country                  180519 non-null  object
 7   Order Region                   180519 non-null  object
 8   Order State                    180519 non-null  object
 9   Order Status                   180519 non-null  object
 10  Order Zipcode                  24840 non-null   float64
 11  shipping date (DateOrders)     180519 non-null  object
 12  Shipping Mode                  180519 non-null  object
dtypes: float64(2), int64(3), object(8)
memory usage: 17.9+ MB
None
```

Figure 2. Dataframe of the DataCo CSV

Encoded the columns for use in the model. Please view Jupyter Notebook for the code and the columns that were encoded. The reduced dataframe for use with the model was created. This code was made earlier but wanted to have a copy of the frame to use.

	Days for shipping (real)	Days for shipment (scheduled)	Customer State	Delivery Status	Shipping Mode
0	3	4	1	3	1
1	5	4	1	0	1
2	4	4	2	1	1
3	3	4	2	3	1
4	2	4	1	3	1

Figure 3. Dataframe of the delivery dataframe

Then there was a check to see if there were any rows that did not have any values in them. There were no rows that did not have values associated with them.

```
Days for shipping (real)         0
Days for shipment (scheduled)    0
Customer State                   0
Delivery Status                  0
Shipping Mode                    0
dtype: int64
```

Figure 4. No NaN found

A correlation matrix was produced to see if there was a reasonable correlation. It is shown below.

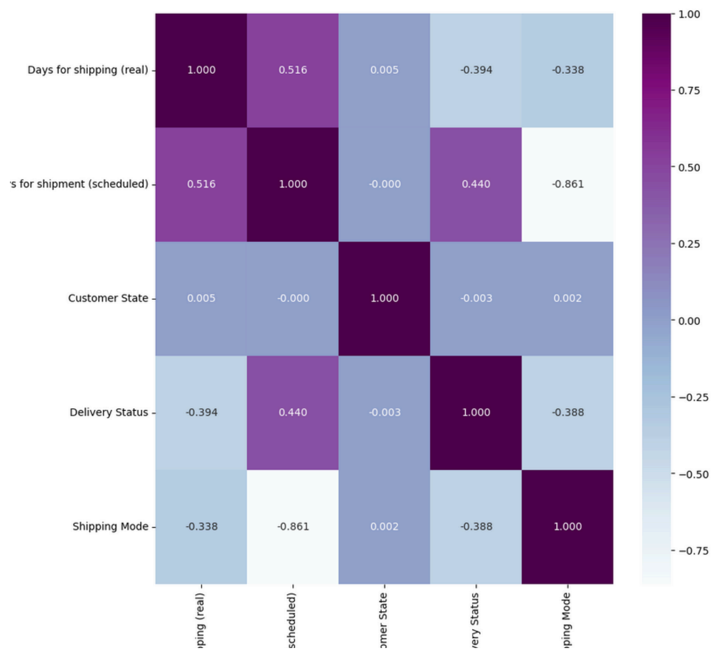

Figure 5. Correlation Matrix

The KNN was created and the result of the model yielded an accuracy of:

```
Accuracy of the KNN model 95.71238643917572
```

An ANN model was also created. Its results proved to be unsatisfactory for this dataset given the inputs and the current setup of the model. It is assumed that ANN[3] could be tuned and offer better results with a different model implementation or possibly a different and more

[3]The implementation of the ANN can be found in the notebook.

extensive data set. The results gained from the KNN model are more than adequate for this dataset, and it is easier to implement and maintain.

Posturing Statement

The model that has been proposed is K-nearest neighbors as it is easy to use. It has the ability to answer the question at hand. This question is, “What caused the delay in delivery?” It is simple but effective in estimating whether or not there will be delay based on the features selected.

Heading Five.

References

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